

Sai Mun Lee

AI SOLUTIONS ARCHITECT | LLM & PLATFORM ENGINEERING

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Summary

Senior software engineer with 8+ years building and shipping production, multi-tenant foundation-model serving platforms at enterprise scale — owning model inference orchestration, tenant isolation, API design, encryption and data governance across 10+ enterprise tenants. Strong in ML-platform and retrieval systems (recommender engines, graph and document retrieval), and building hands-on with LLM/RAG stacks: Claude, OpenAI and Gemini APIs, vector stores and prompt engineering. Spans architecture and delivery, leads teams through zero-to-one, high-ambiguity initiatives, and works fluently with Claude Code and modern gen-AI tooling.

Skills

GenAI & LLM	LLM APIs (Claude, OpenAI, Gemini), RAG Pipelines, Vector Stores (Pinecone, pgvector), Prompt Engineering, Claude Code
ML & Serving	Foundation-Model Serving, Model Inference & Optimization, Recommender Systems, PyTorch
Languages	Python, Go, TypeScript, JavaScript, Bash
Backend & APIs	REST API Design, Microservices, FastAPI, Flask, Django, PostgreSQL, HANA DB, Elasticsearch
Infrastructure	Kubernetes, Docker, Cloud Foundry, Jenkins, Terraform
Cloud	AWS (Solutions Architect Pro, in progress), GCP (Associate Cloud Engineer)
Security	Multi-tenant Isolation, Encryption (at-rest & in-transit)
Practices	Distributed Systems, System Design, ML Platform Architecture, TDD, Code Review
Spoken	English, Chinese

Experience

SAP Asia

Singapore

SENIOR SOFTWARE ENGINEER

Dec 2020 - Present

- Co-architected SAP's Tabular AI platform — a model-agnostic, highly scalable multi-tenant serving layer for the RPT-1 foundation model on Kubernetes (SAP AI Core), targeting 10+ enterprise tenants at 100K users each.
- Owned the end-to-end serving architecture: API contracts, tenant isolation, model inference orchestration and model lifecycle management across the platform.
- Implemented enterprise security — encryption at rest and in transit, and tenant data isolation — for employee data on SAP SuccessFactors.
- Optimized the model inference serving path to improve latency and throughput under concurrent multi-tenant workloads.
- Translated product and stakeholder requirements into technical specifications and architecture decisions, and coordinated delivery across a 15-person cross-functional engineering team.
- Led delivery of Personalized Recommendations, a recommender engine that generated 8B+ recommendations in 2025 for enterprise customers, and architected People Like Me, a service that aggregates customer data into training data for Personalized Recommendations and exposes its own REST API for consuming recommendations.
- Established CI/CD (Jenkins) with unit, integration and e2e test coverage and instrumented the platform with Dynatrace APM and Kibana (correlation-ID tracing), reducing mean-time-to-resolution for production incidents.

TAU Express

Singapore

LEAD SOFTWARE ENGINEER

Apr. 2020 - Dec. 2020

- Led development of a knowledge-management and document-retrieval system enabling analysts to retrieve and correlate insights across cybersecurity documents via graph relationships (Neo4J) and full-text search (ElasticSearch) — retrieval foundations directly relevant to RAG.
- Owned end-to-end delivery: gathered requirements directly from users, designed the system architecture, and facilitated communication between users and the development team.
- Tech: React, Django, Docker, Neo4J, ElasticSearch, Cytoscape.js

SPIRIT Research Center (NTU)

Singapore

RESEARCH ENGINEER

Feb. 2018 - Mar. 2020

- Developed the Intelligent Case Retrieval System (ICRS) for Singapore's Smart Nation initiative — a full-stack document-retrieval application (Django, Vue.js) for efficient retrieval of legal documents.
- Built web crawlers (Selenium, BeautifulSoup) to index Chinese news articles and social media content for document-retrieval pipelines.

Certificates

Dec 2021 **AWS Certified Solutions Architect – Associate**, Amazon Web Services (AWS)

Feb 2022 **GCP Associate Cloud Engineer**, Google Cloud Platform (GCP)

May 2023 **Hashicorp Certified: Terraform Associate (003)**, Hashicorp

AWS Certified Solutions Architect - Professional, Amazon Web Services (AWS)

In Progress

Education

Nanyang Technological University (NTU)

Singapore

B.S. IN COMPUTER SCIENCE (HONOURS) — MACHINE LEARNING & DATA SCIENCE SPECIALISATION

2013 - 2017